SAFETY-FIRST FIELD GUIDE

Poisonous Mushrooms

Recognize risk, document the whole specimen, and act quickly after a possible exposure.

North American research edition

First-edition candidate | Research cutoff 28 August 2026

DIRECT ANSWER

You cannot identify every poisonous mushroom by a shortcut

Safe identification is a species-level conclusion built from multiple features, location, habitat, maturity, and expert comparison. A color, photo, app result, spore print, folk test, or symptom clock cannot certify edibility. [S01][S06] [S11]

POSSIBLE INGESTION IN THE UNITED STATES

1-800-222-1222

Call Poison Help immediately. Do not wait for symptoms and do not induce vomiting. For collapse, seizure, trouble breathing, or inability to awaken, call 911. [S03][S04]

THIS GUIDE CAN

Help you recognize stop patterns and collect evidence for an expert.

IT CANNOT

Declare a wild mushroom edible or replace a poison centre, clinician, or mycologist.

SAFEST DEFAULT

Do not eat a wild mushroom unless a qualified local expert has established its identity.

NAVIGATION

Contents

01	Draw the safety boundary	1	02	Build an evidence packet	4
03	Recognize high-risk patterns	7	04	Treat time as a triage signal	11
A	Notes and sources	15	I	Index of terms	17

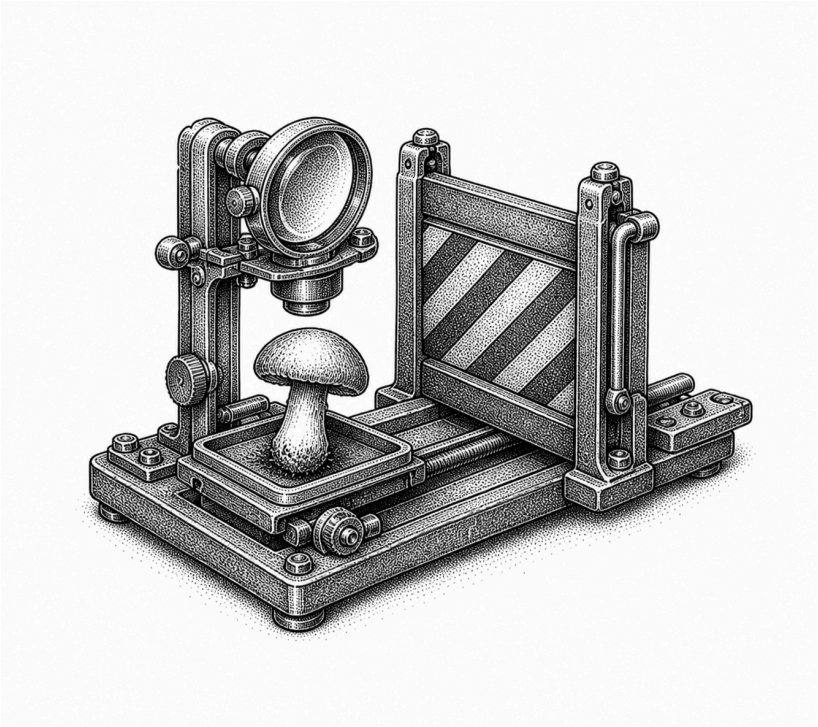
READING RULE

Every visual pattern in this handbook is a reason to stop or seek expert review. Absence of a listed pattern is never evidence that a mushroom is safe.

CHAPTER 01

Draw the Safety Boundary

Identification is a chain of evidence. Edibility is a separate decision with a much higher cost of error.



1.1 Know what an identification does

A mushroom name is a hypothesis until all important features agree with a reliable, regional description and plausible lookalikes have been excluded. For nonexperts, there is no dependable rule that separates every poisonous species from every nonpoisonous species.[S01][S06]

OBSERVATION	Cap, underside, stem, intact base, bruising, odor, substrate, habitat, season, and grouping.
CANDIDATE ID	A species or group that currently fits the evidence. It is not an eating decision.
EXCLUSION WORK	Compare every dangerous lookalike, including immature, damaged, and regional forms.
EDIBILITY DECISION	Requires species certainty, local expertise, condition, preparation knowledge, and individual risk review.

> **STOP CONDITION**
>
> If even one deciding feature is missing, ambiguous, damaged, or inconsistent, the conclusion is *unidentified*. Do not eat it.

Do not reverse the logic

A warning pattern can justify avoidance. Its absence cannot prove safety. A mushroom without a visible ring may still be dangerous; a buried base may hide a volva; cap color and veil remnants can change with age or weather.[S07][S08]

SOURCES [S01] [S06] [S07] [S08]

Author synthesis: this boundary turns taxonomic uncertainty into a safe field decision.

1.2 Reject shortcuts that answer the wrong question

SHORTCUT	WHAT IT MAY SHOW	WHY IT CANNOT PROVE SAFETY
Bright or dull color	A descriptive cap feature	Both toxic and edible mushrooms span many colors.
Animals or insects ate it	Another organism tolerated it	Different species respond differently to toxins.
Silver, onion, peeling, or bruising test	A chemical or physical change	No folk reaction maps reliably to all mushroom toxins.
Cooking, drying, freezing, or washing	Changes texture or some contaminants	Most mushrooms that poison people cannot be made safe by processing.
Spore print alone	Mass spore color	Many species share colors; it is only one character.
Photo app result	A candidate label from visible pixels	It can miss hidden structures, habitat, maturity, and regional lookalikes.

APP EVIDENCE

In a peer-reviewed test of three apps on 78 specimens from poison exposures, the best-performing app identified 49% of specimens correctly and 44% of poisonous specimens correctly. The authors concluded that apps were not reliable enough to exclude a potentially poisonous exposure when used alone.[S11]

Use an app as a notebook or candidate generator, never as permission to eat. A field guide, community comment, or confident AI answer has the same limitation when decisive evidence is absent.

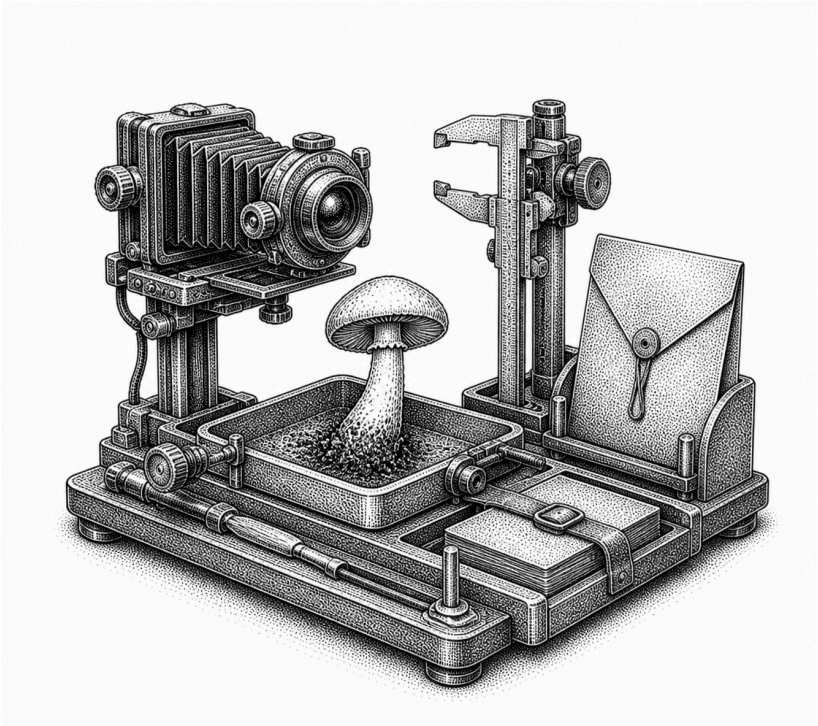
SOURCES [S01] [S07] [S11] [S12]

Author synthesis: the comparison distinguishes information aids from safety authorities.

CHAPTER 02

Build an Evidence Packet

The complete base, underside, habitat, and time record are often more useful than a beautiful cap photo.



2.1 Photograph the evidence before it changes

1. IN PLACE
Show the mushroom, substrate, grouping, and nearby habitat before disturbing it.

2. FULL SIDE
Include a scale and the complete stem from cap to excavated base.

3. CAP
Record shape, surface, margin, color, texture, and veil remnants.

4. UNDERSIDE
Show gills, pores, teeth, ridges, their spacing, and attachment to the stem.

5. BASE
Excavate rather than cut. A bulb or saclike volva may be underground.

6. SECTION
When safe and useful, show internal structure and any color change after cutting.

Record context in plain language

- ☐ Date, general location, weather, and recent rainfall.
- ☐ Substrate: soil, lawn, moss, living tree, log, stump, wood chips, dung, or buried wood.
- ☐ Nearby trees or plants; growing alone, scattered, clustered, or in a ring.
- ☐ Cap and stem measurements, odor without tasting, texture, bruising, and time to color change.
- ☐ Several ages if lawful and safe. Young and old specimens may look materially different.

> **DO NOT TASTE AN UNKNOWN MUSHROOM**
>
> This handbook does not use taste as a field character. Wash hands after handling and prevent children or pets from reaching the specimen.

SOURCES [S05] [S07] [S12]

Author synthesis: the six-view sequence prioritizes the features experts repeatedly request.

2.2 Keep observation separate from emergency evidence

SITUATION	PRESERVE	RECORD	DO NOT DO
Learning or field study	Photos, notes, lawful voucher specimen	All morphology and habitat fields	Do not turn a tentative name into an eating decision
Possible ingestion	Uneaten mushroom, meal scraps, cleaning scraps, photos	Who ate, amount, time, preparation, symptoms	Do not delay the poison-centre call to collect evidence
Mixed collection	Keep each species separate and label each container	Which pieces entered each dish	Do not assume the saved specimen matches every eaten piece

A spore print narrows; it does not clear

Mass spore color can distinguish some lookalikes. Use a mature, fresh cap over contrasting light and dark paper and record the result. A failed print may reflect age or condition. A matching color cannot prove species or edibility.[S07][S12]

FOR EXPERT REVIEW AFTER AN EXPOSURE

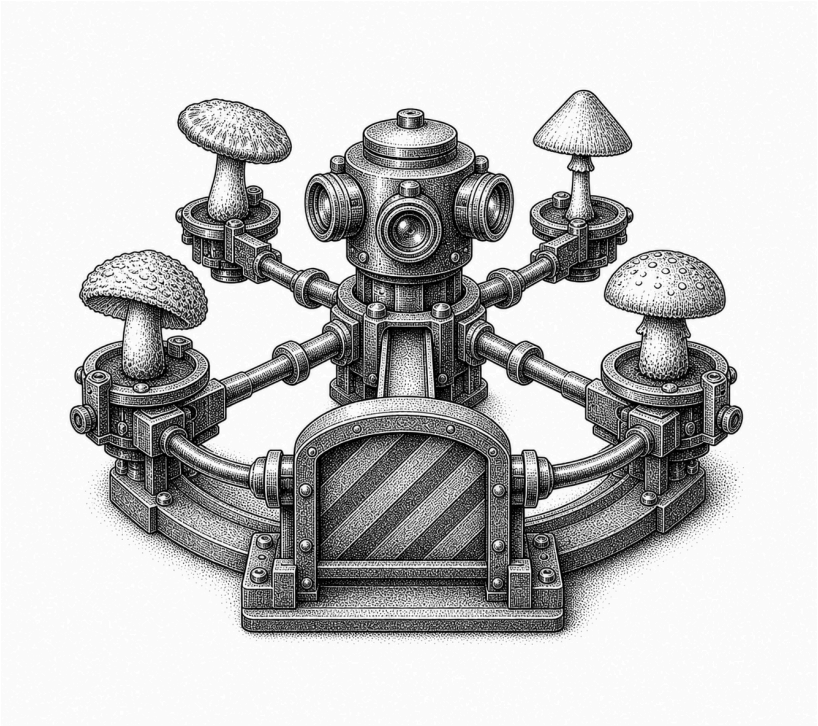
When safely available, keep an intact uncooked specimen or large fragment in a paper or waxed-paper bag, not plastic, and refrigerate it. Photograph the cap, underside, stem, and complete base. Follow the poison centre's directions and do not delay care.[S05][S06]

> **IMPORTANT LIMIT**
>
> Identifying one saved mushroom as nontoxic does not exclude another species in the meal. Mixed collections require separate evidence for every specimen.[S06]

CHAPTER 03

Recognize High-Risk Patterns

A warning constellation tells you to stop. It does not identify every toxic mushroom or prove that anything outside the pattern is safe.



3.1 Treat the Amanita constellation as a hard stop



Death caps and destroying angels are among the amatoxin-producing *Amanita* groups. An important warning combination is pale or white free gills and usually white spores, a fragile stem ring that may be absent with age, a cap that may be white, green, yellow, or brown, and a bulb or saclike volva at the base that may be hidden underground. Any feature may be damaged, lost, or obscured.[S05][S07][S08]

> **ACTION**
>
> Any plausible Amanita constellation means: do not taste, do not cook, keep it away from children and pets, and seek qualified local identification if needed.

Young-button trap. An unopened *Amanita* can resemble a puffball. A true puffball is uniformly solid inside when cut; an Amanita button can show a developing cap and stem. Even this comparison is a warning screen, not an edible clearance.[S08]

SOURCES [S05] [S07] [S08]

Author synthesis: the constellation is deliberately framed as a stop rule, not a species diagnosis.

3.2 Do not let size or familiarity lower the bar

HIGH-RISK PATTERN	FEATURES TO DOCUMENT	WHY IT IS A STOP
Deadly Galerina *Galerina marginata* group	Small brown cap; brown to rusty spores; slight ring or ring zone; on logs, stumps, wood chips, or buried wood	Contains the same amatoxin class found in deadly amanitas and resembles other small brown wood-growers.
Small parasol-like mushrooms Some *Lepiota* and *Pholiotina*	Small stature; gills; ring or veil remnants; cap scales; spore color; habitat	Some species contain amatoxins. The lethal pattern is not confined to large white amanitas.
Green-spored parasol *Chlorophyllum molybdites*	Large whitish scaly lawn cap; ring; mature greenish gills; green spore print; often in rings	Poisonous and commonly confused with edible parasols. Young gills may still look pale.

The "little brown mushroom" problem

Many small brown gilled mushrooms share the same field silhouette. Microscopic characters or DNA may be needed. A cap photo cannot separate all harmless, psychoactive, and amatoxin-containing species. [S08][S10]

> **PRACTICAL RULE**
>
> For a beginner, small brown gilled mushrooms and small parasol-like mushrooms are expert-only. Do not use a spore color or growth substrate as a single-test exception.

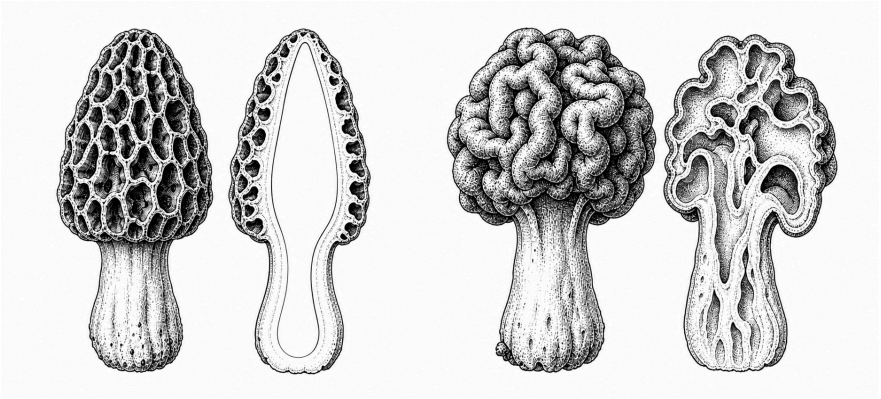
SOURCES [S08] [S10] [S13]

Author synthesis: these groups are combined because their ordinary appearance creates a high-consequence recognition failure.

3.3 Compare structure, then keep the safety limit

True morel pattern

False morel warning



pits and ridges; hollow inside

wrinkled or brainlike; chambered

Michigan guidance treats false morels such as *Gyromitra* as poisonous and not edible. True morels have definite pits and are hollow; false morels may be wrinkled, lobed, waved, brainlike, or chambered. This structural comparison helps flag risk but does not certify a collection as edible.[S09]

Some serious patterns are not visually distinctive

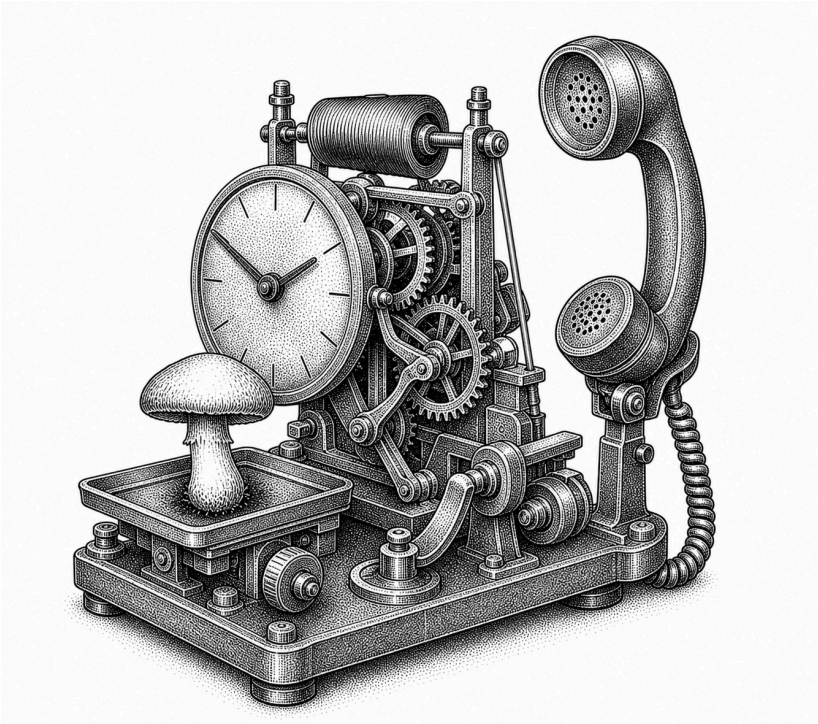
Orellanine-producing *Cortinarius* can cause kidney injury after a delay of days or weeks. The genus is large and difficult. A rusty spore print or web-like veil is not enough to identify a toxic species, and waiting for symptoms is unsafe.[S10][S15]

> **NON-EXHAUSTIVE BY DESIGN**
>
> This chapter highlights common high-consequence traps. It does not cover every toxic species, toxin, region, or life stage.

CHAPTER 04

Treat Time as a Triage Signal

Symptoms can arrive quickly, late, or in misleading phases. The clock helps a poison specialist; it never tells you to wait.



4.1 Use latency to communicate urgency, not to diagnose

15 min to 4 h	Many gastrointestinal irritants cause nausea, cramps, vomiting, or diarrhea. Muscarinic syndromes can add sweating, saliva, tears, small pupils, slow pulse, or breathing difficulty.
30 min to 2 h	Isoxazole-containing mushrooms can cause confusion, visual disturbance, agitation, profound drowsiness, or seizures.
About 4 to 12 h	Gyromitrin poisoning often begins with gastrointestinal illness; severe cases can involve seizures, altered consciousness, liver injury, or blood effects.
6 to 24 h	Amatoxin poisoning classically begins after a quiet interval with abdominal pain, vomiting, and diarrhea. Apparent improvement can precede liver and kidney failure.
36 h to weeks	Orellanine-associated kidney injury can be delayed. Thirst, altered urination, fatigue, nausea, or flank discomfort may emerge long after the meal.

TIMING CAVEAT

Early symptoms do not rule out a severe poisoning. Mixed species, dose, repeated meals, individual susceptibility, and uncertain ingestion time can blur every band.[S02][S10][S15][S16]

Tell the poison specialist the exact eating time, earliest symptom, every later change, and whether more than one species or meal was involved. Do not self-treat from this chart.

SOURCES [S02] [S10] [S15] [S16] [S17]

Author synthesis: ranges are deliberately broad and overlapping because latency is probabilistic, not diagnostic.

4.2 Act first; identify in parallel

UNITED STATES

1-800-222-1222

Call Poison Help immediately after a possible wild-mushroom ingestion. For collapse, seizure, trouble breathing, or inability to awaken, call 911.

01 **Call now.**
Do not wait for symptoms, an app verdict, a social-media reply, or a spore print.

02 **Do not induce vomiting.**
Do not give activated charcoal, milk, food, medicine, or a folk antidote unless the poison specialist or clinician directs it.

03 **Preserve evidence without delaying care.**
Keep mushrooms, meal scraps, cleaning scraps, and photos. Use paper or waxed paper, not plastic, and refrigerate when directed.

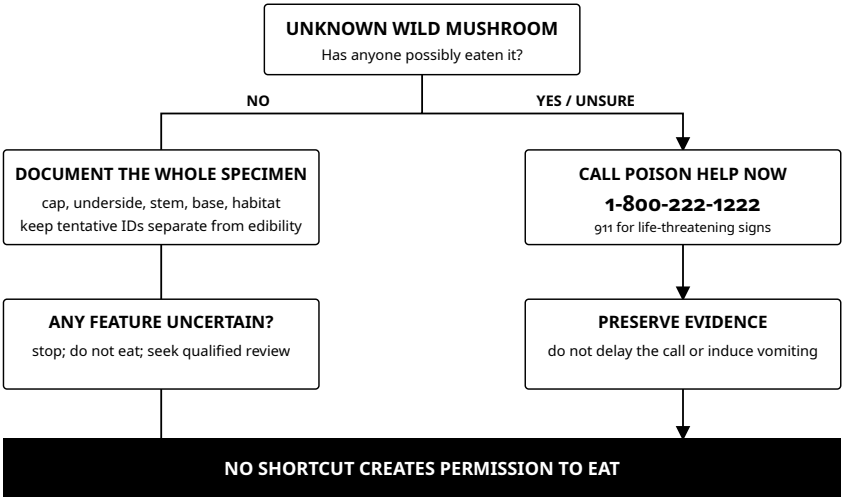
04 **Write the exposure record.**
Person, age, weight if known, amount, time, preparation, symptoms, other foods, and everyone who shared the meal.

05 **Follow the expert pathway.**
The poison centre can coordinate medical care and, when available, qualified mycological identification.

> **OUTSIDE THE UNITED STATES**
>
> Contact your local poisons centre or emergency service. The WHO maintains a world directory, but coverage and hours vary. If severe symptoms are present or no poisons centre is available, use local emergency services or an emergency department.[S14]

SOURCES [S03] [S04] [S05] [S06] [S14]

4.3 Use one conservative decision path



FIELD STUDY

Learn with a regional key, qualified mycologist, and repeated observation across life stages.

FOOD DECISION

Require positive species identification and exclusion of every dangerous lookalike.

EXPOSURE

Call first. Identification supports care but must not delay it.

SOURCES [S01] [S03] [S05] [S06] [S07]

Author synthesis: the flow resolves uncertainty toward the lower-risk action.

APPENDIX A

Notes and Sources

Sources were accessed on 28 August 2026. Stable labels connect claims to direct, inspectable evidence.

[So1] U.S. Food and Drug Administration. Bad Bug Book, 2nd Edition: Mushroom Toxins. 2012. Federal overview; intentionally not a comprehensive clinical reference.

[So2] Brandecker KJ et al. CDC MMWR. Amanita Species Mushroom Poisonings - Northern California, November 2025-March 2026. 28 May 2026.

[So3] U.S. HHS Health Resources and Services Administration. Poison Help: What You Can Do. Current emergency guidance.

[So4] California Poison Control System. Plants and Mushrooms. Current first-aid and prevention guidance.

[So5] California Department of Public Health. Poisonous Wild Mushrooms. Current through May 2026.

[So6] Moss M. Utah Poison Control Center. Mushroom Ingestion - Diagnosis and Management. August 2021.

[So7] Missouri Department of Conservation. A Guide to Missouri's Edible and Poisonous Mushrooms. 2021. Regional field guide.

[So8] U.S. Forest Service, Alaska Region. Mushrooms of the National Forests in Alaska. Regional field guide; publication date not stated in the retrieved copy.

Authority and limitations are recorded in the companion claim ledger. Regional sources support only the morphology and geography they actually describe.

APPENDIX A CONTINUED

Clinical and Identification Sources

[S09] Michigan Department of Natural Resources. Morel Mushroom Identification. Regional true-morel and false-morel comparison.

[S10] North American Mycological Association. Mushroom Poisoning Syndromes. Specialist, non-exhaustive syndrome overview.

[S11] Hodgson SE et al. A comparison of the accuracy of mushroom identification applications using digital photographs. *Clinical Toxicology*. 2023;61(3):166-172.

[S12] Iowa State University Extension and Outreach. Tips for Collecting and Identifying Mushrooms. Field-documentation guidance; no edibility recommendations.

[S13] University of Florida IFAS Extension. The Green-Spore Poison Parasol Mushroom, Chlorophyllum molybdites. January 2016; reviewed April 2025.

[S14] World Health Organization. World Directory of Poisons Centres. Directory dated 2023.

[S15] Wennig R et al. Mushroom Poisoning. *Deutsches Arzteblatt International*. 2020;117:701-708.

[S16] Graeme KA. Mycetism: A Review of the Recent Literature. *Journal of Medical Toxicology*. 2014;10:173-189.

[S17] Vohra V et al. A 19-year longitudinal assessment of gyromitrin-containing mushroom poisonings in Michigan. *Toxicon*. 2024;247:107825.

Material limitations

No local mycologist or medical toxicologist reviewed this candidate. Species coverage is intentionally incomplete. Morphology changes with age, damage, weather, and region. Clinical latency ranges overlap and cannot diagnose a toxin. The app study tested three products on 78 Australian exposure specimens; it supports an app-alone warning, not a permanent ranking of all identification tools.

INDEX

Index of Terms

A

AI and photo apps - 3
Amanita - 7, 8, 12
Amatoxin - 7-9, 12
Apparent recovery - 12

B

Base, excavated - 4-6, 8
Bruising - 5

C

Cap color - 2, 3, 8
Chlorophyllum molybdites - 9
Cooking - 3
Cortinarius - 10, 12

E

Emergency response - ii, 11-14
Evidence packet - 4-6
Expert review - ii, 1-6, 14

F

False morels - 7, 10
Folk tests - 3

G

Galerina - 7, 9
Gills - 4, 5, 8, 9
Gyromitrin - 10, 12

P

Poison Help - ii, 11, 13, 14
Preserving specimens - 6, 13

R

Ring - 4, 8, 9

S

Spore print - 2, 3, 6, 8, 9
Symptom timing - 11, 12

V

Volva - 2, 4, 5, 8

W

WHO poisons-centre directory - 13, 16
Warning constellation - iii, 7-10

FINAL SAFETY RULE

An unknown wild mushroom stays unknown until qualified evidence resolves it. If ingestion might have occurred, call first and identify in parallel.

This first-edition candidate has no prior approved edition for comparison.